by using vice for simple operations. No other special devices are required to hold and guide the tool for simple operations, but for some special work and production works, the tool must be guided and the job to be held in correct position by of special device. Such special device is called Jigs and Fixtures.

We must understand that the device which holds the job and guides the tool is called jig and the device which holds the job only in correct position is called fixture.

Purpose of Jigs and Fixtures

1. It is to provide repeatability and interchangeability
2. To get accuracy in the manufacturing products
3. To provide interchangeability for alternative parts to another one

JIG

Jig is a special device used to hold the work pieces and guide and Control the tools used in the machining process.

Important Parts of Jig

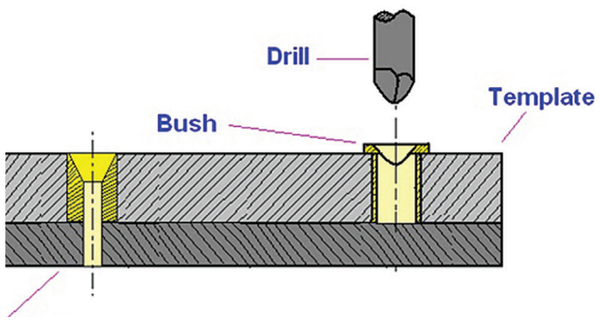
1. Jig Body
2. Jig Feet
3. Jig bush
4. Bush Plate (or) Jig Plate
5. Locators
6. Clamps

Types of Jig

1. Plate Jig
2. Angle Jig
3. Channel Jig
4. Box Jig
5. Diameter Jig
6. Leaf Jig
7. Vice Jig
8. Ring Jig
9. Indexing Jig

Plate Jig: This is the simplest type of Jig. It is a simple plate made to the shape and size of the work piece, with the required number of holes made on it. This plate Jig is mounted on the work piece and held rigidly. Then the work is drilled rapidly by passing the drill bit through the hole on the plate Jig.

Angle Plate Jig: It resembles as an angle plate. One side of the angle plate is fitted on the machine table. A plate is fixed parallel to the base, on the top of the angle plate. Drill bushes are attached to guide the drills on that plate. The work is



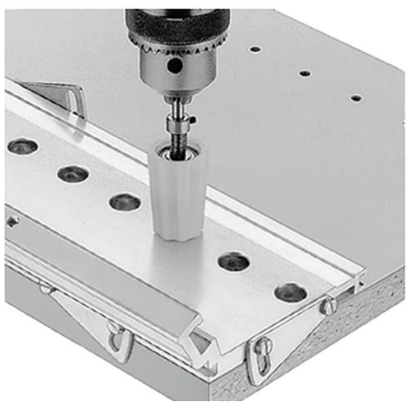


Plate Jig

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